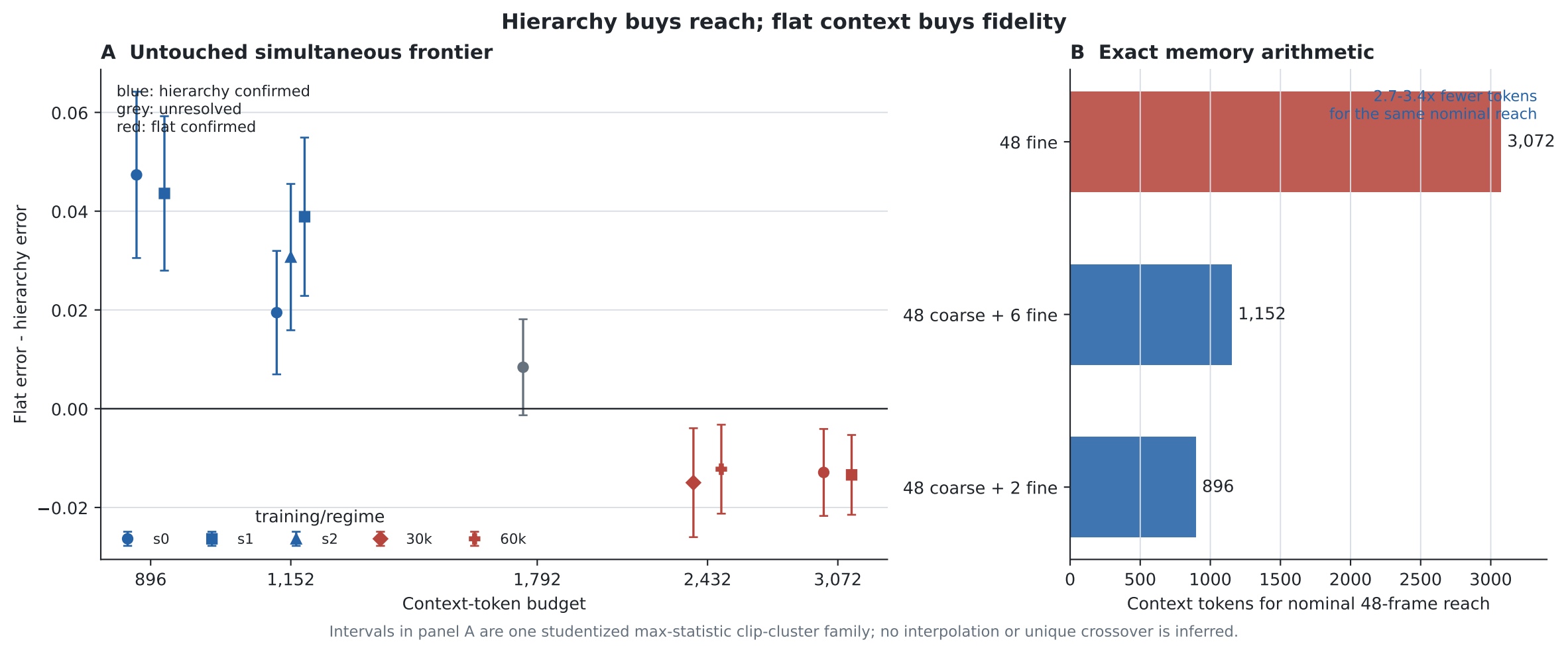
Hierarchy buys reach; flat context buys fidelity
A Untouched simultaneous frontier
blue: hierarchy confirmed
grey: unresolved
red: flat confirmed
Flat error - hierarchy error
0.06
0.04
0.02
0.00
−0.02
training/regime
s0
s1
s2
30k
60k
896
1,152
1,792
2,432
3,072
Context-token budget
B Exact memory arithmetic
2.7-3.4x fewer tokens
for the same nominal reach
48 fine
3,072
48 coarse + 6 fine
1,152
48 coarse + 2 fine
896
0
500
1000
1500
2000
2500
3000
Context tokens for nominal 48-frame reach
Intervals in panel A are one studentized max-statistic clip-cluster family; no interpolation or unique crossover is inferred.